

Speech Emotion Recognition using Deep Learning

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Abstract

This report provides a detailed analysis of our deep learning project, which focuses on speech emotion recognition using the Toronto Emotional Speech Set (TESS) dataset. The primary objective of this project is to explore the application of state-of-the-art machine learning techniques to classify emotions from speech signals. By leveraging the Wav2Vec2 model, we demonstrate significant advancements in processing and understanding audio data for emotional intelligence in machines. This report discusses the methodology, implementation, results, and potential areas for future improvement.

1 Introduction

Speech emotion recognition (SER) is an essential component of human-computer interaction, enabling systems to interpret and respond to human emotions effectively. Recognizing emotions from speech has applications in diverse fields, including healthcare, customer service, and entertainment. Our project focuses on classifying emotions such as happiness, sadness, anger, and fear using deep learning techniques.

The Toronto Emotional Speech Set (TESS) dataset was chosen for its comprehensive coverage of emotional speech samples. This dataset consists of audio recordings by two actresses, emulating seven distinct emotions. The inclusion of high-quality recordings ensures robustness in model training and evaluation.

1.1 Project Objective

The primary aim of this project is to develop a deep learning model capable of accurately classifying speech emotions. Key objectives include:

- Preprocessing the audio data to extract meaningful features.
- Fine-tuning the Wav2Vec2 model for optimal performance on the TESS dataset.
- Evaluating model performance using appropriate metrics and visualizations.

2 Methodology

2.1 Data Collection and Preprocessing

The TESS dataset was sourced from Kaggle and extracted for analysis. Initial exploration of the dataset involved loading audio files and inspecting their properties, such as duration and sampling rate. Using `librosa`, key preprocessing steps included:

- **Resampling:** Ensuring uniformity in audio sample rates for compatibility with the Wav2Vec2 model.
- **Noise Reduction:** Applying filters to minimize background noise and enhance signal clarity.
- **Feature Extraction:** Generating spectrograms and mel-frequency cepstral coefficients (MFCCs) to capture essential audio features.
- **Data Splitting:** Dividing the dataset into training, validation, and testing subsets with an 80-10-10 ratio.

2.2 Model Architecture

The Wav2Vec2 model, developed by Facebook AI, serves as the backbone of our project. This transformer-based architecture is well-suited for speech processing tasks, offering:

- Pretrained feature extraction layers that encode raw audio into rich representations.

- Fine-tuning capabilities for specific tasks, such as emotion classification.
- A robust design that handles variable-length audio sequences effectively.

The model was adapted for sequence classification by adding a linear classification head, enabling it to predict one of the seven emotion categories.

2.3 Training and Evaluation

The training process involved fine-tuning the Wav2Vec2 model using the `Trainer` API. Hyperparameters were meticulously selected to ensure efficient convergence:

- **Learning Rate:** $5e-5$, optimized for gradual weight adjustments.
- **Batch Size:** 16, balancing memory constraints and computational efficiency.
- **Epochs:** 3, sufficient for model convergence on the TESS dataset.
- **Evaluation Strategy:** Validation metrics computed at the end of each epoch.

Training metrics included:

- Runtime statistics, highlighting processing speed and resource utilization.
- Loss curves, which showed consistent improvement over epochs, confirming effective model training.
- Confusion matrices, identifying strengths and weaknesses in emotion classification.

3 Results and Discussion

3.1 Training Performance

The final training output is summarized as follows:

```
TrainOutput(global_step=390, training_loss=0.649, metrics={  
    'train_runtime': 166.7927,  
    'train_samples_per_second': 37.412,  
    'train_steps_per_second': 2.338,  
    'total_flos': 1.133e+17,  
    'train_loss': 0.649,  
    'epoch': 3.0  
})
```

The training loss of 0.649 indicates the model’s strong learning capability. Processing over 37 samples per second demonstrates the model’s efficiency, leveraging GPU acceleration effectively. This suggests that the model was well-optimized for the dataset and training parameters.

3.2 Evaluation Results

The evaluation results for the test dataset are as follows:

```
{'eval_loss': 0.1836, 'eval_accuracy': 0.9981, 'eval_precision': 0.9981,  
 'eval_recall': 0.9981, 'eval_f1': 0.9981, 'eval_runtime': 3.8449,  
 'eval_samples_per_second': 135.243, 'eval_steps_per_second': 8.583, 'epoch': 3.0}
```

The model achieved an exceptionally high accuracy of 99.81%, with precision and recall metrics reflecting a similarly robust performance. The evaluation runtime of 3.84 seconds highlights the model’s computational efficiency during testing. With a real-world test result of 2 out of 2 correctly classified samples, the model’s practical applicability is strongly validated.

The loss curves from training showed a consistent decrease, indicating that the model effectively minimized errors and avoided overfitting. These trends confirm the suitability of the selected hyperparameters and the model’s capability to generalize well on unseen data.

4 Conclusion

Our project successfully implemented a deep learning framework for speech emotion recognition using the TESS dataset. The fine-tuned Wav2Vec2 model demonstrated high accuracy and efficient resource utilization. Key achievements include:

- Effective preprocessing techniques that enhanced audio quality.
- A robust model architecture that accurately classified emotions.

- Consistent training progress as evidenced by loss reduction trends.

Future work will explore:

- Expanding the dataset to include more diverse speakers and emotions.
- Incorporating ensemble methods to further enhance classification accuracy.
- Real-time deployment of the model in practical applications, such as virtual assistants.